Caffe2-rs

Caffe2-rs is a Rust crate that provides an operator network library for the Rust ecosystem. The library is based on the flexible and robust Caffe2 C++ library and aims to provide high performance, high modularity, and ease of integration into all future systems written in Rust.

The goal of this project is to finish translating the implementations of each operator and supporting crates. Although much of the code is still in need of polish, the interfaces are roughly in place. The translation task is now parallelized. To ensure functionality, the unit tests from the C++ codebase have been included.

In the near future, this library could serve as a reference model for PyTorch, as well as a tool for the Rust ecosystem. The project owes its DNA to the original authors of the Caffe2 C++ library, without whom this work would not exist.

Contributing

Contributions to this project are welcome and encouraged. If you're interested in contributing, please take a look at the contribution guidelines.

License

Caffe2-rs has a BSD-style license, as found in the LICENSE file.

README Disclaimer

The README files for each crate are being generated via a varying length, medium depth conversation with chatgpt.

I'm not sure what everybody else thinks about this, but it seems pretty awesome to me.

In certain circumstances I have been able to drill down into some depth to find out what the bot knows about the different algorithms.

The only thing to watch out for is this: the README files are only *probabilistically correct*

I repeat: you can't rely on them totally.

They serve more as guidelines, than hard and fast rules.

Sometimes, I left information from these conversations in the README even though I *knew* it wasn't entirely accurate. (although it may be accurate one day)

For example, sometimes, the bot generated example usage code based on its knowledge of just a few symbols from the crate. There is probably no way this generated code works out of the box. As of March 6, 2023, I haven't tried all of it.

To me, at least, this information is still useful.

This is because it represents the bot's best guess on how this crate might be used.

It is useful to me to know how the bot might try and want to use the crate, even if the APIs it is asking for dont actually exist yet (for example).

In another example, sometimes I ask the bot to do some numerical simulations to calculate performance behavior. Please don't blindly trust these numbers. I'm not totally sure they're correct even though they look like a pretty good start.

In other words, please take the information in the README files only for what it actually is, and what it actually represents.

Please consider them a guideline and not a contract.

I provide this information in case it is interesting, and so you yourself don't have to spend the sleep wake cycles asking the bot if you don't want to.

Nothin but love,

-klebs

BTW if you find glaring errors in these files, plz feel free to correct them. Correcting the possible mistakes is totally welcome.

Also note:

As of 3/06/2023 I am currently still in the process of documenting and reorganizing some details.
If you are already checking this crate out, first of all, you are awesome, hello! but second of all, please know that the structure of this workspace is somewhat in flux.

Acknowledgments

The author of this project would like to thank the original authors of the Caffe2 C++ library for their work, as well as the Rust community for their support.